



Differential Calculus: From Practice to Theory by Eugene Boman & Robert Rogers is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/), except where otherwise noted.

© 2023 Eugene Boman & Robert Rogers

ISBN: 978-1-942341-95-6

Cover art: “Problem of Delos Constructed from a Solution by Isaac Newton (Arithmetica Universalis)” by Crockett Johnson, courtesy of Ruth Krauss in memory of Crockett Johnson, Division of Medicine and Science, National Museum of American History, Smithsonian Institution

Published by Milne Open Textbooks, Milne Library
State University of New York at Geneseo,
Geneseo, NY 14454

A Note on the Cover

On the front cover you see the images of four central figures in the development of differential calculus. These are certainly not all of the players involved in this story but they represent a transition from the practical aspects (Isaac Newton and Gottfried Leibniz) to the foundational underpinnings (George Berkeley and Joseph Lagrange). On the following page and back cover you see a more complete (but certainly not exhaustive) timeline of some of the people involved and their contributions.

In 1707, in his book [Arithmetica Universalis](#), Isaac Newton presented a solution of the ancient [Problem of Delos](#) (Doubling the Cube). In 1970 the artist Crockett Johnson used Newton's diagram as the inspiration for the painting [Problem of Delos Constructed from a Solution by Isaac Newton](#) which is reproduced (with the permission of the Smithsonian Institute) as the background image on the covers of this book. The original problem stipulates that the solution be constructed using only a compass and straightedge i.e., using the Geometry of Euclid. It is now known that the Problem of Delos cannot be solved using only Euclidean Geometry. Newton's solution is correct but it requires a very small extension of Euclid's Geometry.

Differential Calculus Chronology

Birth of Galileo	1564	
	1638	Discourses Concerning Two New Sciences — Galileo
Death of Galileo; Birth of Newton	1642	
Birth of Leibniz	1646	
Birth of L'Hôpital	1661	
	1665	Method of Fluxions — Newton
Birth of Bernoulli	1667	
	1684	A New Method for Maxima and Minima calculus — Leibniz
Birth of Taylor; Birth of Berkeley	1685	
	1687	Principia Mathematica — Newton
	1691	L'Hôpital's rule — Bernoulli
	1696	Analyse des Infiniment Petits – L'Hôpital
Death of L'Hôpital	1704	
Birth of Euler	1706	
	1715	Taylor series – Taylor
Death of Leibniz	1716	
Birth of Agnesi	1718	
Death of Newton	1727	
Death of Taylor	1731	
	1734	The Analyst — Berkeley
Birth of Lagrange	1736	
Death of Bernoulli	1748	Analytical Institutions — Agnesi
Death of Berkeley	1753	
	1772	Fonction Dérivée — Lagrange
Birth of Bolzano	1781	
Death of Euler	1783	
Birth of Cauchy	1789	
	1797	Mean-value theorem — Lagrange
Death of Agnesi	1799	
Death of Lagrange	1813	
Birth of Weierstrass	1815	
	1817	Continuity and the intermediate-value theorem — Bolzano
	1823	Limit definition of a derivative — Cauchy
Death of Bolzano	1848	
Death of Cauchy	1857	
	1861	Extreme-value theorem — Weierstrass
	1874	Formal definition of the limit of a function — Weierstrass
Death of Weierstrass	1897	
Birth of Robinson	1918	
	1960	Rigorization of infinitesimals — Robinson
Death of Robinson	1974	

Acknowledgments

The authors wish first and foremost to express our gratitude to the students who were subjected to various rough drafts of this textbook in a classroom setting. Our intention was always to write a student-centered textbook. To whatever extent we have succeeded in achieving that goal the insights and suggestions of these students were an invaluable help and we are very grateful to them for that.

Second, we wish to thank Dr. H. Joseph Straight, Distinguished Service Professor at SUNY, Fredonia for agreeing to use some of the earlier drafts in his classes. With grace and professionalism, he guided his students through, and sometimes around, the poor organization, murky examples, and innumerable typos of these early versions, simultaneously giving us precious insights and suggestions for improvement. His comments were always on point and we are deeply appreciative of all of his efforts.

We also wish to thank Allison Brown at Milne Open Textbooks for her patience with us. We are sure that she grew very tired of getting emails from us saying that we were almost done only to find that we still had a lot more polishing to do. Through it all her support was unwavering and we are deeply indebted to her for it.

Last, but certainly not least, we wish to thank our copy editor, Dr. Margo Boman, for all of her help in getting us to the current version of this book. When we employed her in this capacity, we thought that we were nearly done. It turned out that we were sadly mistaken. Margo's attention to detail and her pedagogical expertise were instrumental in transforming a collection of stories and problems into a much more coherent and polished narrative. It was more work than both she and we anticipated, but it was worth it and we are deeply beholden to her. Future teachers and students using this book will benefit enormously from her contribution.

Finally, any errors that might still appear in this text are the fault, and sole responsibility, of the authors.